

ABHAY ANJANEYAVALLABH KOKA

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EDUCATION

KL University Hyderabad

Bachelor of Technology in Artificial Intelligence and Data Science with a CGPA of 9.4

2022 - Expected June 2026

Hyderabad, India

Specialization: Cloud and Edge Computing

PROJECTS

Diabetic Retinopathy Classification using Deep Learning (*Python, TensorFlow, Scikit-learn*) — *Ongoing*

- Developing a deep learning model to classify stages of Diabetic Retinopathy from retinal fundus images using TensorFlow and Keras, with Scikit-learn for data preprocessing and evaluation.

Car Insurance Claim Data Analysis using XGBoost and MLflow (*XGBoost, Python, MLflow, Machine Learning*)

- Built an end-to-end machine learning pipeline using XGBoost to predict car insurance claims, incorporating 5-fold cross-validation and early stopping for improved generalization.
- Performed hyperparameter tuning and evaluated model performance using AUC, F1-score, and confusion matrix.
- Leveraged MLflow for experiment tracking, model versioning, and artifact logging.
- Codebase and documentation: **GitHub Repository**

Route Optimization Web App: ACO vs Classical TSP (*Django, ACO, TSP, Floyd-Warshall, Python, Render*)

- Developed a Django-based web application to visualize and compare Ant Colony Optimization with the classical TSP algorithm.
- Used the Floyd-Warshall algorithm to precompute all-pairs shortest paths for efficient travel distance calculations.
- Simulated pheromone-based adaptive routing in ACO and compared performance with brute-force TSP based on distance and time.
- Visualized the route evolution and convergence behavior of ACO over multiple iterations using dynamic plotting.
- Deployed the application on Render for real-time access and testing. **Live Demo**

Sentiment Analysis of Text (*Python, Scikit-learn, Django, NLP*)

- Built a machine learning model to classify text sentiment as positive, neutral, or negative using natural language processing techniques. **Demo**

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, R
- **Technologies/Frameworks:** Django, HTML/CSS, SQL

PROFESSIONAL CERTIFICATIONS

- PCAP - Certified Associate Python Programmer
- PCEP - Certified Entry-Level Python Programmer
- Red Hat Certified Enterprise Application Developer

OTHER CERTIFICATES

- Introduction to Cybersecurity (Cisco Netacad)
- Data Science Simulation - British Airways
- Internshala - Business Analytics
- Operating Systems and You (Coursera x Google)
- Technology Job Simulation - Deloitte Australia
- ChatGPT Prompt Engineering for Developers

LANGUAGES

- English, Hindi, Telugu